

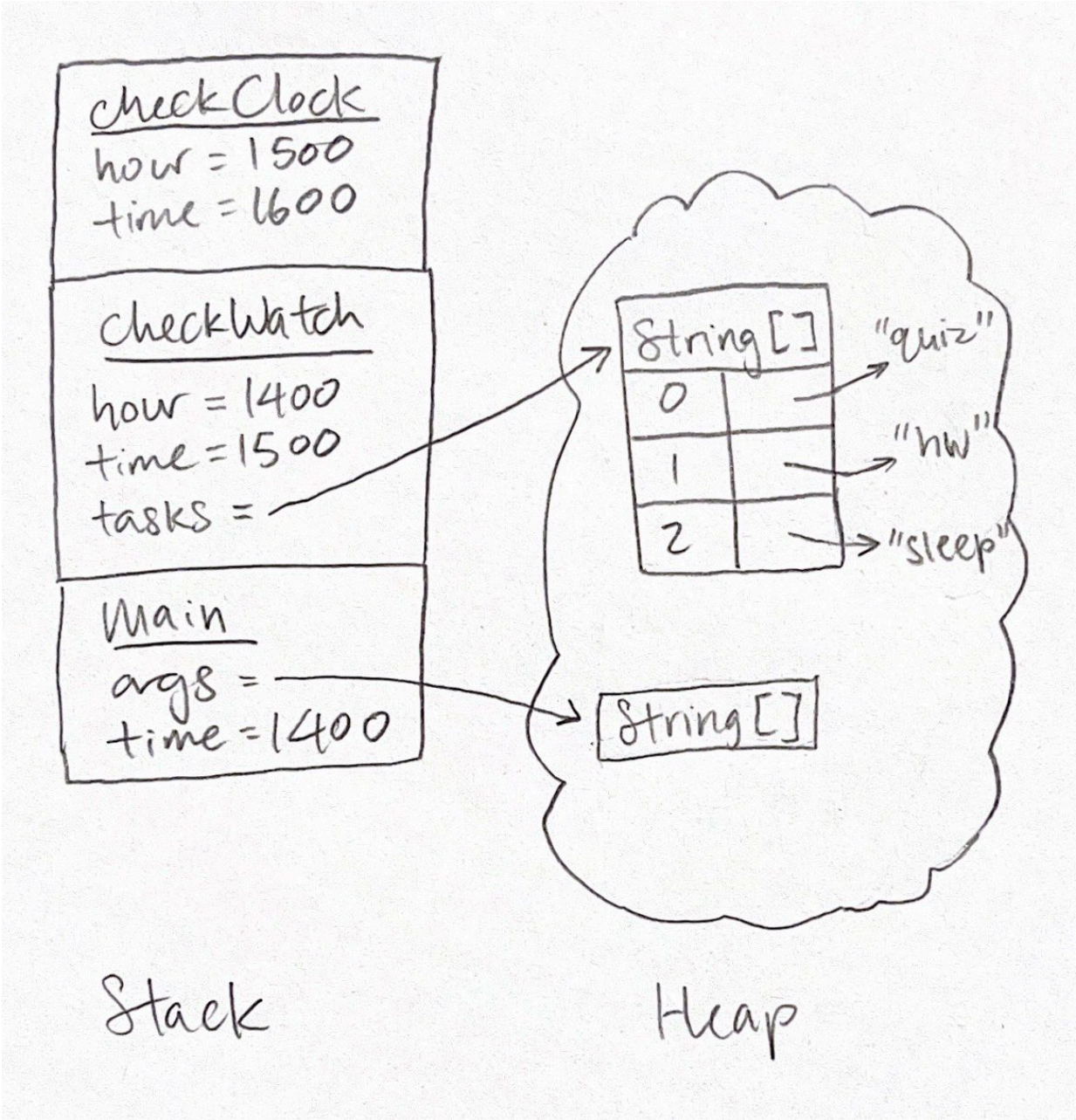
Student  
Steven Chen

Total Points  
27.5 / 33 pts

Question 1  
(no title)

4 / 5 pts

+ 5 pts



✓ + 4 pts Minor errors

+ 3 pts Some errors

+ 2 pts Major errors, such as:

- No stack frames
- Not creating a new time and hour variable for each method call

+ 0 pts Incorrect

Question 2  
(no title)

2 / 2 pts

✓ + 2 pts Instance fields, class fields, constructors, instance methods, class methods

+ 1.5 pts Minor errors

+ 1 pt Major errors

+ 0 pts Incorrect

Question 3

(no title)

1 / 1 pt

✓ + 1 pt Object

+ 0 pts Incorrect

Question 4

(no title)

1 / 1 pt

✓ + 1 pt null

+ 0 pts Incorrect

Question 5

(no title)

2 / 2 pts

✓ + 2 pts To separate the the abstract behavior of a data type from its concrete implementation

+ 1 pt Partially correct

+ 0 pts Incorrect

Question 6

(no title)

1 / 1 pt

✓ + 1 pt ∞

+ 0 pts Incorrect

Question 7

(no title)

2 / 2 pts

✓ + 2 pts Its class, any interfaces its class implements, its class' superclasses

+ 1.33 pts Two correct

+ 0.67 pts One correct

+ 0 pts None correct

Question 8

(no title)

1 / 1 pt

✓ + 1 pt False

+ 0 pts Incorrect

9.1 (no title)

3 / 4 pts

+ 4 pts

```
public interface Table {
    String getMaterial();
    void setMaterial(String material);
}
```

✓ + 3 pts Minor errors or style issues

+ 2 pts Some errors or style issues

+ 1 pt Major errors or style issues

+ 0 pts Incorrect

9.2 (no title)

5 / 6 pts

+ 6 pts

```
public class TableImpl implements Table {
    private String _material;

    public TableImpl(String material) {
        _material = material;
    }

    public String getMaterial() {
        return _material;
    }

    public void setMaterial(String material) {
        _material = material;
    }
}
```

✓ + 5 pts Minor errors or style issues

+ 3 pts Some errors or style issues

+ 2 pts Major errors or style issues

+ 0 pts Incorrect

9.3 (no title)

2 / 3 pts

+ 3 pts Two examples of tests:

```
@Test
public void getTest() {
    Table table = new TableImpl( material: "wood");
    assertEquals( expected: "wood", table.getMaterial());
}

@Test
public void setTest() {
    Table table = new TableImpl( material: "wood");
    table.setMaterial("metal");
    assertEquals( expected: "metal", table.getMaterial());
}
```

✓ + 2 pts Minor errors or style issues

+ 1 pt Major errors or style issues

+ 0 pts Incorrect

Question 10

(no title)

1 / 1 pt

✓ + 1 pt  $O(n \log n)$

+ 0 pts Incorrect

Question 11

(no title)

1 / 1 pt

✓ + 1 pt  $O(n^2)$

+ 0 pts Incorrect

Question 12

(no title)

1.5 / 3 pts

+ 3 pts  $O(xn^2), O(n^2 + xn^2)$

+ 2 pts Mostly correct answer/work

✓ + 1.5 pts Some correct answer/work

+ 0 pts Incorrect

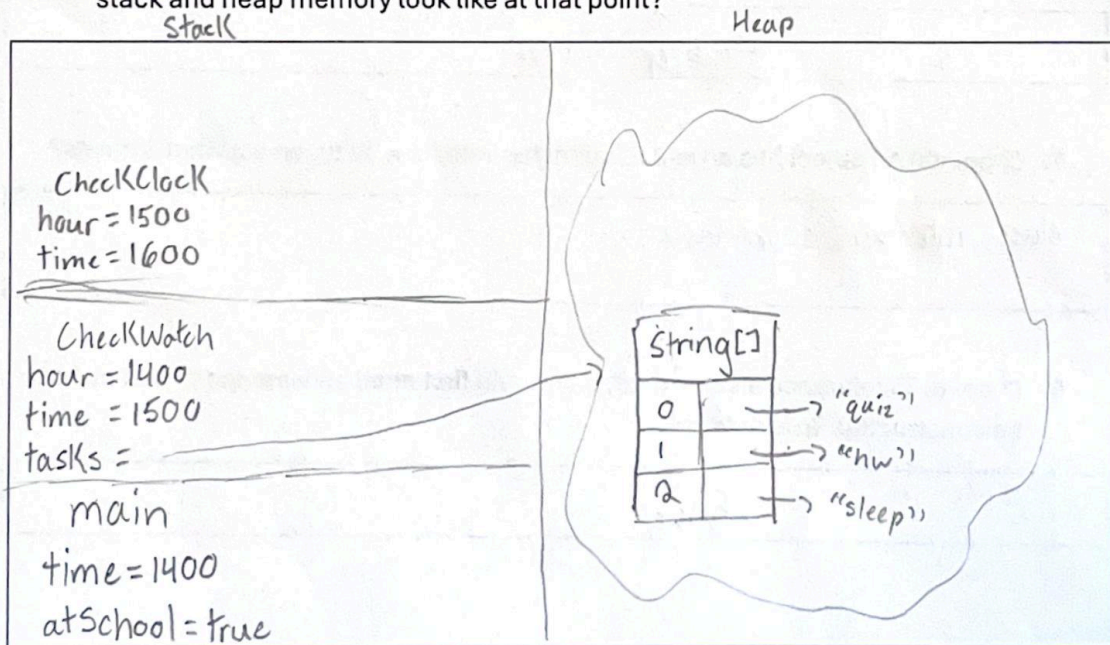
## COMP 210 Q2

Name: Steven Chen

Consider the following code.

```
1 public static void main(String[] args) {  
2     int time = 1400;  
3     checkWatch(time);  
4     boolean atSchool = true;  
5 }  
6  
7 public static void checkWatch(int hour) {  
8     int time = 1500;  
9     String[] tasks = {"quiz", "hw", "sleep"};  
10    checkClock(time);  
11 }  
12  
13 public static void checkClock(int hour) {  
14     int time = 1600;  
15     System.out.println(time);  
16 }
```

1. (5 points) If we put a breakpoint on line 15 and ran the debugger, what would the stack and heap memory look like at that point?



output: 1600

Steven Chen

2. (2 points) What are the 5 components of a class in order from top to bottom?

Instance Field → Class Field → Constructor → Instance Method → Class Method

3. (1 point) What is the universal reference type?

Object

4. (1 point) What is the universal reference value?

null

5. (2 points) Describe the purpose of ADTs in a sentence.

A contract that promises what the behavior will be without showcasing implementation, similar to cars having brakes and gas but actual implementation varies.

6. (1 point) A single ADT can have how many implementations?

1

However Many

7. (2 points) An object has an is-a relationship with (can be typed as) what 3 things?

Class, interfaces, superclasses

8. (1 point) To reference a static field, one would first need an instance of that class to be constructed. True or false?

False

10. (1 point) What is the average case time complexity of quick sort?

$O(n \log n)$

11. (1 point) What is the worst case time complexity of quick sort?

$O(n^2)$

Consider the following method.

```
public static int myMethod(int n, int x) {
    int counter = 0;

    if (n % 2 == 0) {
        for(int i = 0; i < n; i++) {  $\leftarrow O(n)$  }
        for(int j = i; j > 0; j--) {  $\leftarrow O(n)$  }
        counter++;
    }
    } else {
        for(int k = x; k < x * 3; k++) {  $O(k^3)$  }
        for(int m = 5; m < n * n; m++) {  $O(k^2) = O(k^5)$  }
        counter++;
    }
}

return counter;
}
```

$k^2 \cdot k^3 = k^5$

12. (3 points) What is the time complexity in terms of n and x? Show your work in the code above.

$O(n^2 + k^5)$

(/●ㄣ●)/\*:°◇

Great job!!!!!!

(/●ㄣ●)/\*:°◇

9. I want to create an ADT called Table. For its behavior, I want to be able to retrieve and modify the material it's made from.
- a. (4 points) Write a simple interface for Table. Remember to use good style and naming conventions.

```
public interface Table {
    String getMaterial();
    void setMaterial();
}
```

- b. (6 points) Write a simple class that implements your interface. Remember to use good style and naming conventions.

```
public class TableImp implements Table {
    private String material;

    public Table(String material) {
        this.material = material;
    }

    public String getMaterial() {
        return material;
    }

    public void setMaterial(String newMaterial) {
        material = newMaterial;
    }
}
```

- c. (3 points) Write a simple JUnit test (method) with one assertion to test one of your interface methods. Remember to include the annotation.  
**Hint:** Declare an interface type object inside the method.

```
@Test
public void testGetMaterial() {
    TableImp newTable = new Table("wood");
    assertEquals("wood", newTable.getMaterial());
}
```

Steven Chen

Nongraded question (you will be given time to answer it at the end):

Any updates on suggestions, feedback, or opinions since last time? (Example topics: lectures, assignments, quizzes, content, activities, policies, etc.)

Quiz time was a bit tight.

Autograder for assignments so we can resubmit and fix mistakes

